Graphic Era Hill University
DEHRADUN • BHIMTAL • HALDWANI

Hack- o – Holic 4.0

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Project Title:- **TrueNFT : AI + Blockchain-Powered NFT Authenticity Verification & Fraud Detection System**

Team Name: Apex
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College Name: Graphic Era Deemed to be University

Problem Statement

Problem Statement:
The global NFT market suffers from rampant fraud due to lack of verification, harming creators and buyers alike.
TrueNFT combats this with AI and blockchain to detect fakes, verify ownership, and blacklist bad actors.

Detailed description about the problem:

- NFT market to hit $232B by 2030, yet $100M+ lost to scams in 2021–22; U.S. Treasury labels NFTs “highly susceptible” to fraud.
- 80% of OpenSea’s NFTs are plagiarized or spam; Marketplaces like Cent shut down due to rampant fraud.
- No marketplace offers AI-powered authenticity checks — all fraud responses are manual and ineffective.
- Research (Pham 2023, Pungila 2022) confirms AI + blockchain can detect plagiarism and restore trust.
- Creators fear theft, buyers risk fake assets — real financial harm erodes confidence in NFTs.

Proposed Solution

Solution Statement:

- AI Detection: Uses OpenAI's CLIP + pHash to spot fake NFTs with 90%+ accuracy vs. verified originals.
- Ownership Check: Confirms creator legitimacy via blockchain history using Moralis API.
- Fraud Scoring: Combines AI + blockchain data for real-time 0–100 fraud score (Authentic/Suspicious/Fake).
- Blacklist: Flags bad actors permanently on Polygon's immutable, transparent smart contract.
- All-in-One Trust: Only platform merging detection, verification, scoring, and blacklisting in real time.

Proposed Solution

Detailed Solution Description:

- 4-in-1 Pipeline: Real-time AI scanning, ownership check, fraud scoring, and blacklisting in one system.
- Ownership Proof: Uses Moralis + Alchemy to confirm if seller is the original NFT creator.
- AI Detection: OpenAI CLIP + pHash spots fakes with 90%+ accuracy (validated by Pham et al. 2023).
- Fraud Score & Blacklist: Generates 0–100 score; flags bad actors permanently on Polygon via Solidity.
- Plug-and-Play: API-first design lets marketplaces integrate instantly; supports multi-chain, aligns with SDGs 8,9,16

Technical Stack

Technologies to Be Used:

Step 1 — Input: User submits an NFT's contract address + token ID via the TrueNFT web dashboard.

Step 2 — AI Scan: CLIP model + pHash engine compares the NFT image against the verified originals database and generates a visual similarity score.

Step 3 — Chain Verify: Moralis API fetches the complete on-chain mint history and ownership trail of the submitted NFT.

Step 4 — Fraud Score: Both signals feed into the weighted Fraud Scoring Engine, producing a 0–100 authenticity score.

Step 5 — Result & Action: NFT is classified as Authentic, Suspicious, or Fake — if fake, the seller's wallet is automatically pushed to the Solidity blacklist smart contract on Polygon.

Technical Stack

Detailed Description:

- **Hardware:** Standard laptop/desktop (minimum 8GB RAM, GPU optional for AI inference) — no specialised hardware required as AI processing is handled server-side via cloud deployment. Any internet-connected device with a browser can access TrueNFT, making it hardware-agnostic and widely accessible.
- **Software:**
- v**Frontend:** React.js + Tailwind CSS — for building a responsive, real-time web dashboard with wallet connectivity via MetaMask/WalletConnect.
- v**AI Engine:** Python (FastAPI) + OpenAI CLIP Model + ImageHash (pHash) library — for semantic image embedding, similarity computation, and near-duplicate detection.
- v**Smart Contracts:** Solidity — for writing the blacklist registry and ownership verification contracts, compiled and tested using Hardhat framework.
- v**Backend:** Node.js + Express.js — for API routing, connecting the frontend to the AI engine and blockchain layers.
- v**Programming Languages:** Python, JavaScript, Solidity

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System

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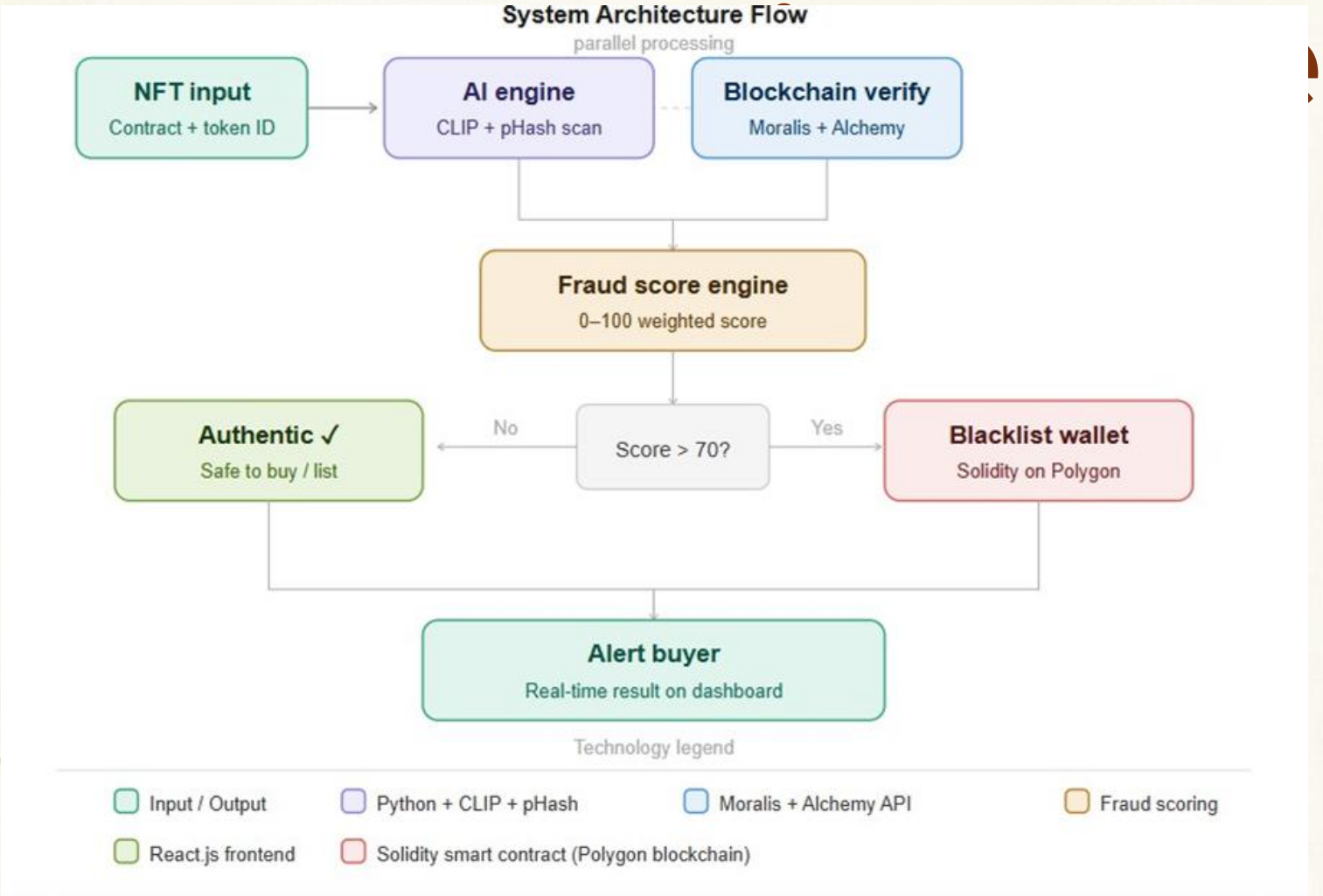

Real World Application and Challenges of the solution you provide

Real-World Applications :

- Marketplaces & Artists: Integrate TrueNFT API to auto-scan listings or register original NFTs — block fakes before they go live.
- Buyers & Brands: Get instant authenticity reports via contract address; monitor branded NFTs for unauthorized copies in real time.
- DAOs & Law Enforcement: Verify voting NFTs to prevent manipulation; use on-chain blacklist as tamper-proof evidence in IP/fraud cases.

Challenges & Mitigations :

- Data & Accuracy: Build verified NFT database incrementally; add human review for borderline AI scores to reduce false positives.
- Tech & Scale: Start with Ethereum/Polygon; cache results and batch API calls to manage cost and multi-chain complexity.
- Fairness & Legal: Use DAO-governed appeals for blacklists; position as transparency tool — flag, don't enforce — to navigate global legal ambiguity.